

Program	B. Tech CSE: All Batches	Semester	I
Course	Physics for Computer Engineers	Course Code	PHYS1036
Session	July-December, 2025	Topic	Unit I: Lasers & Fibre Optics

Key Points and Formulae

- LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. It is a device to produce an intense, monochromatic, collimated and highly coherent beam of light. The operation of a LASER depends upon the phenomenon of stimulated emission.
- Population inversion is a necessary criterion for laser action; if N_2 and N_1 are the number of atoms in excited and ground states respectively, then population inversion may be mathematically written as $N_2/N_1 > 1$. If E_2 and E_1 are the energies of the excited and ground states respectively, then $N_2/N_1 = \exp \left[- \left(\frac{E_2 - E_1}{KT} \right) \right]$, where K is the Boltzmann constant.
- Angular Spread: $d\theta = 1.22\lambda/D$, λ is the wavelength and D is the diameter of the front mirror.
- Areal spread $= (f \cdot d\theta)^2$, where f is the focal length and $d\theta$ is the angular spread.
- The ratio of stimulated and spontaneous emissions is given by $S_1 = \frac{1}{\exp(h\nu/KT)-1}$; ν is the frequency of laser light.

MULTIPLE-CHOICE QUESTIONS

1. He-Ne Laser is a
 - (a) 3-level Laser
 - (b) 4-level Laser
 - (c) 2-level Laser
 - (d) None of these
2. What color of light does the Ruby LASER emit?
 - (a) Blue
 - (b) Yellow
 - (c) White
 - (d) Red
3. Out of the following, which one is a two-level LASER?
 - (a) CO₂ LASER
 - (b) He-Ne LASER
 - (c) Ruby LASER
 - (d) Semiconductor LASER

4. The directionality of the LASER beam is measured by
- (a) Visibility of Interference fringes
 - (b) Size and aperture of the laser source
 - (c) the divergence angle of the beam with distance from the source
 - (d) Nature of the lasing medium

TUTORIAL SHEET

1. Calculate the power per unit area delivered by a LASER pulse of energy 4×10^{-3} Joule and pulse length as 10^{-9} sec when the pulse is focused on target to a very small spot of radius 1.5×10^{-5} m. **Ans: $5.7 \times 10^{15} \text{ W/m}^2$**
2. A 0.1 watt LASER beam with an aperture of 5.0 mm emits light of wavelength 6943 Å. Calculate the areal spread and intensity of the image when the beam is focused with the lens having a focal length of 100 mm. **Ans: $9.02 \times 10^{-14} \text{ m}^2$; $1.11 \times 10^{12} \text{ W/m}^2$**
3. If level 1 and level 2 are separated by an energy $E_2 - E_1$ such that the corresponding transition frequency falls in the middle of the visible range, Calculate the ratio of the population of the two levels in thermal equilibrium at room temperature. **Ans: 1.357×10^{-38}**
4. The coherence length of sodium light is 2.945×10^{-2} m and its wavelength is 5890 Å. Calculate (i) the number of oscillations corresponding to coherence length and (ii) the coherence time. **Ans: number of oscillations $n = 5 \times 10^4$, Coherence Time: $9.82 \times 10^{-11} \text{ sec}$**
5. Find the intensity of a laser beam of 100 mW power and having a diameter of 1.3 mm. Assume the intensity to be uniform. **Ans: 75.37 kW/m^2**
6. Give the conditions for LASER action to take place.
7. Calculate the population ratio of two states in He- Ne Laser that produces light of wavelength 6000 Å at 300 K? **Ans: e^{-80}**
8. List out few industrial applications of lasers.
9. Calculate the ratio of stimulated to spontaneous emission at a temperature of 300°C for Sodium D line. **Ans: 5.64×10^{-19}**
10. The first line of the principal series of sodium D-Line at 5890 Å corresponds to a transition from the first excited state to the ground state. What is the energy (in eV) of the first excited state? **Ans: 2.1063 eV**

HOLOGRAPHY

Key Points and formula

- Holography was invented by English Physicist Denis Gabor in 1947.
- Light waves reflected from an object are recorded. This photograph record is called a hologram.
- Holography is a two-stage process: In the first stage, a hologram is recorded in the form of an interference pattern. In the second stage, the hologram acts as a diffraction grating for the reconstruction beam, and the image of the object is reconstructed from the hologram.
- Several applications like data storage, security, and holographic interferometry, etc. are few applications.

MULTIPLE-CHOICE QUESTIONS

1. Holography records ____ of light coming from an object.
(a) Intensities (b) Phases
(c) Both (a) and (b) (d) None of the above
2. Through Holography we can produce __dimensional image of objects
(a) One (b) Two
(c) Three (d) None of the above
3. Holography Produces ____
(a) Real images (b) Virtual
(c) Both (a) and (b) (d) None of the above
4. A recorded holographic plate contains information in the form of ____ Pattern
(a) Interference (b) Diffraction
(c) Both (a) and (b) (d) None of the above.
5. In holography, the interference pattern is produced from ____
(a) Object Beam (b) Reference beam
(c) Both a) and b) (d) None of the above

SUBJECTIVE QUESTIONS

1. Mention any two differences between a hologram and a photograph.
2. What is holography? Describe the construction and reconstruction methods of a hologram?
3. Elaborate on the principle of light used in holography.
4. Explain some engineering applications of Holography.
5. What is Holographic data storage? Explain some non-optical applications of holography.

FIBER OPTICS

Key Points and formula

- If n_1 be the refractive index of the core and n_2 , be the refractive index of the cladding, then for total internal reflection
 - $n_1 > n_2$, i.e., the refractive index of the core should be greater than the cladding
 - The angle of incidence on the core-cladding interface (w.r.t. vertical) should be greater than the critical angle.
- When $\theta = \theta_c$, where θ is the angle of incidence and θ_c is the critical angle, the ray traverses along with the interface (i.e. the angle of refraction is 90°).
- The light should travel from the core to the cladding.
- Fiber optics is technology related to the transportation of optical energy (light energy) in guiding media specifically glass fibers.
- Based on transmission properties and the structure, we can categorize optical fibers as single-mode fiber or multimode fiber. The typical diameter of the core of the single-mode fiber is $10\mu m$ and that of multimode fiber ranges from $50\mu m$ to $100\mu m$.

MULTIPLE-CHOICE QUESTIONS

1. The basic material for glass fiber is

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|--------------|--------------|
| (a) B_2O_3 | (b) P_2O_5 |
| (c) GeO_2 | (d) SiO_2 |

2. Propagation of light through a fiber core depends upon the phenomenon of

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|------------------|-------------------------------|
| (a) Interference | (b) Diffraction |
| (c) Polarization | (d) Total internal reflection |

3. The numerical aperture of an optical fiber depends upon

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|---------------------------|--------------------|
| (a) Core refractive index | (b) critical angle |
| (c) Both (a) and (b) | (d) none of these |

4. The main energy losses in optical fibers consists of

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|-------------------|----------------------|
| (a) Material loss | (b) scattering loss |
| (c) Bending loss | (d) All of the above |

5. The angle of acceptance is maximum for a fiber of

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|-----------------------------------|------------------------------------|
| (a) the critical angle is minimum | (b) the critical angle is zero |
| (c) the critical angle is maximum | (d) the critical angle is negative |

TUTORIAL SHEET

1. A step-index fiber has a numerical aperture of 0.26, a core refractive index of 1.5, and a core diameter of 100 μm . Calculate the refractive index of the cladding. **Ans: 1.4772**
2. Calculate the numerical aperture of an optical fiber whose core and cladding are made of materials of refractive indices 1.6 and 1.5, respectively. **Ans: $NA = 0.56$, $\theta_o = 34.1^\circ$**
3. The numerical aperture of an optical fiber is 0.25 and the relative refractive index difference is 0.02. Determine the refractive indices of the core and cladding of a fiber.
4. Calculate the numerical aperture and the acceptance angle of optical fiber from the following data: the refractive index of the core is 1.55 and the refractive index of the cladding is 1.50.
Ans: 0.391; 23°
5. The refractive indices for the core and cladding for a step-index fiber are 1.52 and 1.41, respectively. Calculate i) critical angle ii) numerical aperture iii) the maximum incidence angle.
Ans: $\theta_c = 68.1^\circ$, $NA = 0.568$, $\theta_o = 34.6^\circ$
6. A step index fibre has a core diameter of 0.05 mm and numerical aperture of 0.22 at a wavelength of 8500 $\text{\AA}$. What are the normalised frequency (V_n) and number of modes guided in the core?
Ans: Number of modes = 825
7. How many modes can propagate in a step-index fibre with a core diameter as 40 μm , if the refractive indices of its core and cladding are 1.461 and 1.456, respectively, and the light of wavelength is 8500 $\text{\AA}$?
Ans: Number of modes = 159
8. A signal of power 5 μW exists just inside the entrance of 0.1 km long fibre. Calculate the absorption coefficient of the fibre if the power inside the fibre be 1 μW . **Ans: $\alpha = 70 \text{ dB/km}$**
9. An optical fibre cable 3.0 km long is made up of three 1.0 km length spliced together. The losses due to each length and splice are respectively 5 dB and 1.0 dB. What would be out put power if the input power is 5 mW?
Ans: $P_0 = 0.080 \text{ mW}$